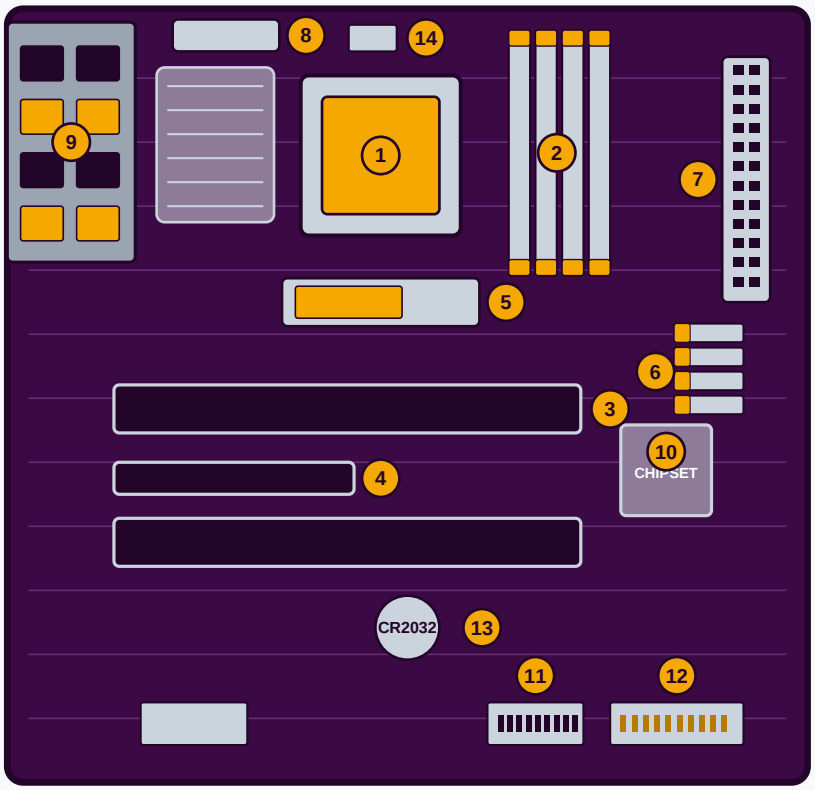
MOTHERBOARD ASSIGNMENT

Slots, ports, CPU sockets and how the main board works

Inside this file: **1.** Labeled ATX motherboard diagram **2.** Three CPU socket types **3.** CPU and chipset, and the Spotify example **4.** Identifying ports and slots with circles

1 Labeled diagram of an ATX motherboard



1. **CPU socket** — Holds the processor
2. **RAM (DIMM) slots** — Memory sticks go here
3. **PCIe x16 slot** — Graphics card
4. **PCIe x1 slot** — Wi-Fi, sound, LAN cards
5. **M.2 slot** — Small, fast SSD
6. **SATA ports** — HDD / SSD / DVD cables
7. **24-pin ATX power** — Main power from PSU
8. **8-pin CPU power** — Extra power for CPU
9. **Rear I/O panel** — USB, HDMI, LAN, audio
10. **Chipset** — Traffic manager for devices
11. **USB header** — Front-panel USB ports
12. **Front panel header** — Power / reset buttons, LEDs
13. **CMOS battery** — Keeps BIOS time and settings
14. **CPU fan header** — Power for the CPU cooler

Figure 1: ATX motherboard with 14 numbered slots, ports and connectors (simplified, not to scale).

A few simple terms

ATX is the most common motherboard size. It measures about 305 mm x 244 mm and has up to 7 expansion slots, so it fits standard desktop cases. Smaller sizes (micro-ATX and mini-ITX) have fewer slots.

Word	Meaning (easy language)
Slot	A long connector on the board where you push in a part. Examples: RAM slots, PCIe slots, M.2 slot.
Port	A socket you plug a cable or device into from outside. Examples: USB, HDMI, LAN and audio on the rear I/O panel, and SATA ports.
Header	A group of metal pins on the board. Cables from the case (front USB, power button, fans) are attached here.
Socket	The square holder for the CPU. Each CPU fits only a matching socket type.

Remember: The numbers in Figure 1 match the key on the right. For example, **1** is the CPU socket and **2** are the RAM slots. Real boards look a little different, but the main parts are always there.

2 Three popular CPU socket types

A **CPU socket** is the holder on the motherboard where the processor sits. The CPU and the socket must be the same type, or the CPU will not fit or work. **LGA** means the pins are on the motherboard (Intel). **PGA** means the pins are on the CPU itself (AMD AM4).

Socket	Brand	Type	Example compatible CPU	Good to know
LGA1151	Intel	LGA, 1151 contacts	**Intel Core i5-8400** (8th gen)	Used for 6th to 9th gen CPUs, but 6th/7th gen and 8th/9th gen need different motherboard chipsets (100/200 series vs 300 series).
AM4	AMD	PGA, 1331 pins	**AMD Ryzen 5 5600X**	Used for many Ryzen generations. Older boards may need a BIOS update for newer Ryzen CPUs.
LGA1200	Intel	LGA, 1200 contacts	**Intel Core i5-10400F** (10th gen)	Works with 10th and 11th gen Intel CPUs on 400/500 series boards. Looks like LGA1151 but is not interchangeable.

Newer sockets: Current PCs use **AM5** (AMD, e.g. Ryzen 7 7800X3D) and **LGA1700** (Intel, e.g. Core i5-12400F). Always check the CPU support list on the motherboard maker's website before buying a CPU.

3 Role of the CPU and the chipset, and how they work with Spotify

CPU (Central Processing Unit)

The CPU is the **brain** of the computer. It reads instructions from programs and carries them out: running apps, doing calculations and making decisions. It works together with RAM, which gives it quick access to the data it needs right now.

Chipset

The chipset is like the **traffic police and receptionist** of the motherboard. It manages the data flow between the CPU and the many other devices such as USB ports, SATA drives, extra M.2 SSDs, network and sound. It decides how many USB ports, SATA ports and expansion slots the board can offer. On modern boards some of these jobs are inside the CPU, but the chipset still handles the extra devices.

What happens when you open Spotify

1. You click the Spotify icon.
The mouse or keyboard sends the signal through a USB port. The chipset passes it to the CPU.

2. The app is loaded.
The CPU asks the SSD or HDD for the Spotify files. The chipset helps move the data, and it is placed in RAM so the CPU can reach it quickly.

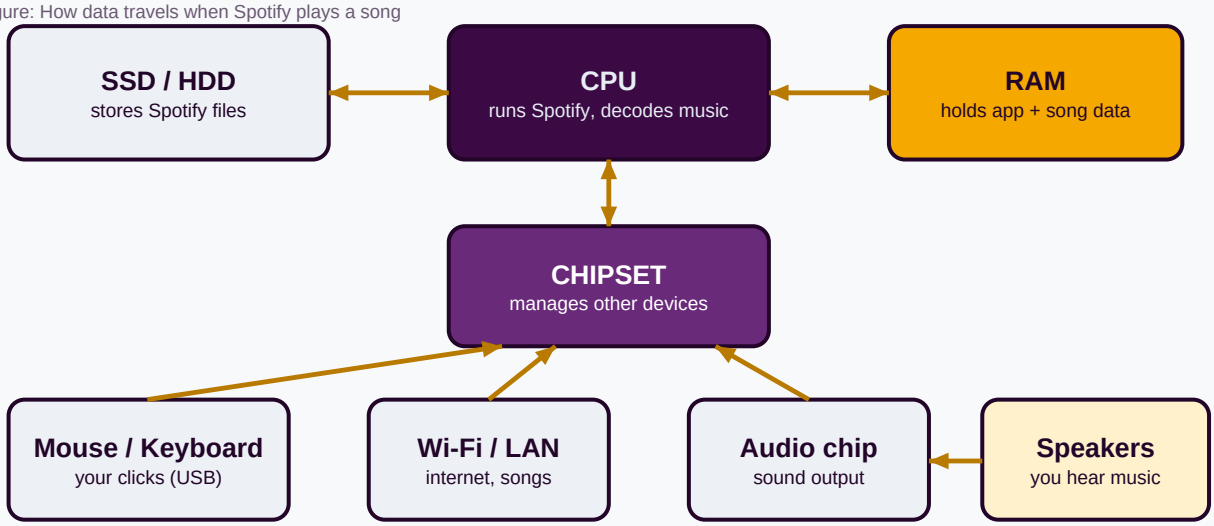
3. The CPU runs the app.
It runs Spotify's code, builds the window you see (with help from the GPU) and waits for your next click.

4. You press play on a song.
Spotify connects to the internet through the Wi-Fi or LAN chip. The chipset passes the incoming song data to the CPU and RAM.

5. The CPU decodes the music.
The song arrives in compressed form. The CPU turns it into sound data, and RAM keeps a few seconds buffered so playback does not stutter.

6. You hear the music.
The sound data goes through the chipset to the audio chip, then to your speakers or headphones.

Figure: How data travels when Spotify plays a song



In one line: The **CPU** does the thinking and running, and the **chipset** helps all the other parts send data to the CPU and receive it back.

4 Identifying ports and slots on a motherboard (circled and labeled)

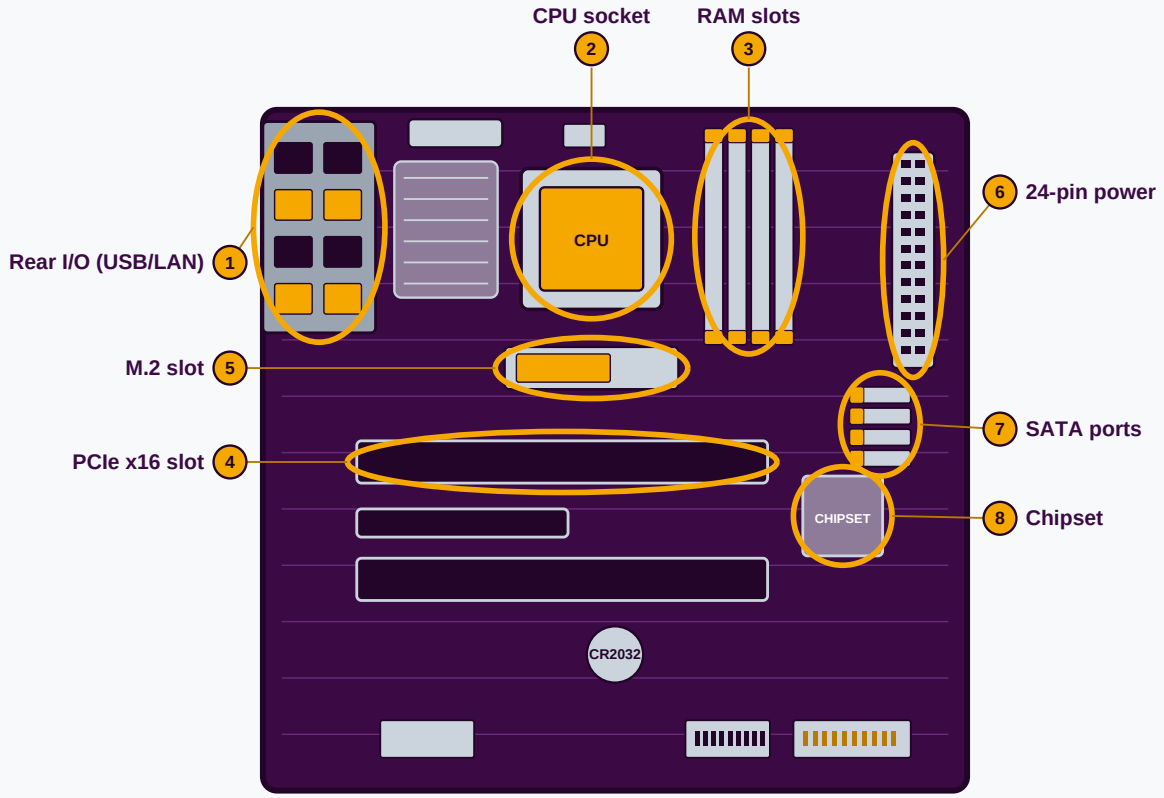


Figure 2: Eight parts circled and labeled on a motherboard top view. This is a drawn illustration, not a real photograph.

No	Part	How to recognise it	What it does
1	**Rear I/O (USB/LAN)**	Metal block at the back edge with many ports.	Connects keyboard, mouse, monitor, internet cable and audio.
2	**CPU socket**	Square holder with a metal cover or lever.	Holds the processor.
3	**RAM slots**	Long thin slots with clips at the ends.	Hold the memory sticks.
4	**PCIe x16 slot**	The longest slot, often with a metal-reinforced edge.	Holds the graphics card.
5	**M.2 slot**	Short slot with a small screw, near the CPU.	Holds a small, very fast SSD.
6	**24-pin power**	Biggest plastic connector, on the right edge.	Brings power from the PSU to the whole board.
7	**SATA ports**	Small L-shaped connectors in a row.	Connect SATA SSDs, HDDs and DVD drives.
8	**Chipset**	Square chip under a heatsink, lower right.	Manages extra USB, SATA and PCIe devices.

Using your own real photo: Take a top-down photo of an open PC, or use a free motherboard image from the internet. Draw a circle around each part and write its name next to it. Use the "How to recognise it" column above as your guide, and write the source of the photo under it.